

MUGUR-AKSY, Russian FederationWMO: **362780**Lat: **50.382N**Lon: **90.434E**Elev: **6070**StdP: **11.75**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-21.0	-32.0	1.1	-25.6	-26.9	1.5	-19.9	19.0	5.9	17.8	5.6	3.2	280	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	76.6	53.6	73.3	52.9	70.3	51.9	55.7	70.4	54.5	68.9	53.4	67.1	8.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.7	68.7	58.9	48.8	64.1	59.1	47.0	59.9	58.8	26.2	70.3	25.4	69.3	24.6	67.2	62.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.4	16.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	-4.1	2.6	20.2	33.3	43.5	56.2	58.5	54.8	43.4	29.9	12.9	0.8		
	DBStd	23.14	8.48	9.97	9.39	8.36	7.95	5.79	4.33	4.72	7.90	9.02	10.12	9.13		
	HDD50	8176	1676	1328	925	501	230	17	1	14	221	624	1114	1525		
	HDD65	12976	2141	1748	1390	951	667	266	206	317	648	1088	1564	1990		
	CDD50	680	0	0	0	0	28	203	265	161	23	0	0	0		
	CDD65	7	0	0	0	0	0	3	4	0	0	0	0	0		
	CDH74	210	0	0	0	0	2	83	107	18	0	0	0	0		
	CDH80	11	0	0	0	0	0	4	7	0	0	0	0	0		
	CDH80	11	0	0	0	0	0	4	7	0	0	0	0	0		
(14) Wind	WSAvg	6.4	4.2	5.8	6.4	7.4	8.2	6.4	7.0	6.0	6.5	6.5	6.9	5.5		
(15) Precipitation	PrecAvg	5.3	0.1	0.1	0.2	0.2	0.3	0.9	1.6	1.1	0.4	0.2	0.2	0.1		
	PrecMax	12.7	0.4	0.5	0.9	0.5	0.7	2.0	6.2	4.8	1.7	0.5	1.0	0.4		
	PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.1	0.4	0.1	0.0	0.0	0.0	0.0		
	PrecStd	2.2	0.1	0.1	0.2	0.1	0.2	0.6	1.1	1.0	0.4	0.2	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	33.4	46.4	60.6	72.7	79.7	80.4	76.6	70.3	57.2	41.3	26.7		
		MCWB	17.4	27.0	34.9	45.0	50.2	54.4	54.8	53.1	49.8	42.2	34.9	23.7		
	2%	DB	16.7	27.3	42.8	54.8	67.2	76.6	77.4	72.4	65.6	53.1	35.8	22.4		
		MCWB	13.6	22.5	32.3	40.8	47.8	53.2	54.5	52.7	48.3	39.9	30.8	19.2		
	5%	DB	12.5	22.7	39.3	50.6	63.0	73.0	73.5	69.5	61.9	49.1	30.4	18.2		
		MCWB	10.0	18.4	30.6	38.1	45.3	52.5	53.5	51.9	46.6	37.7	26.0	15.4		
	10%	DB	9.0	18.6	35.6	47.3	59.0	69.7	70.2	66.6	57.7	44.2	26.8	14.9		
		MCWB	6.8	15.1	28.4	36.2	43.9	51.1	52.7	51.0	45.5	35.1	22.7	12.6		
	0.4%	WB	17.6	27.6	36.6	46.9	51.9	57.2	57.7	57.0	51.8	43.2	36.2	24.3		
		MCDB	20.8	32.8	44.1	58.1	69.3	71.3	71.4	67.1	66.5	55.4	40.8	26.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.8	22.5	33.1	42.2	49.0	54.9	56.0	54.9	49.4	40.6	30.9	19.4		
		MCDB	16.7	26.8	41.5	54.0	65.1	70.3	71.6	67.1	63.1	51.2	35.6	22.5		
	5%	WB	10.1	18.6	31.1	39.1	46.7	53.7	54.9	53.5	47.6	37.9	26.1	15.5		
		MCDB	12.4	22.6	38.5	48.6	61.2	68.9	70.1	66.1	60.5	48.3	30.0	18.1		
	10%	WB	6.8	15.1	28.7	36.9	44.2	52.4	53.8	52.1	45.6	35.3	22.9	12.8		
		MCDB	8.9	18.7	34.9	46.1	57.5	67.5	67.6	64.2	57.4	44.0	26.6	15.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.8	21.0	21.6	20.6	22.2	22.7	21.1	20.6	21.9	19.2	17.6	16.4		
		MCDBR	19.8	23.8	24.8	25.4	28.5	28.2	27.0	26.0	27.8	25.9	21.4	16.0		
		MCWBR	17.4	19.3	16.6	15.2	14.6	12.3	11.3	11.4	14.2	15.8	16.7	13.9		
	5% WB	MCDBR	20.1	23.0	22.5	23.0	27.3	25.3	22.9	22.1	25.4	24.4	21.2	15.5		
		MCWBR	17.8	18.6	15.3	14.7	15.6	12.1	11.5	11.4	13.6	15.7	16.6	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.224	0.248	0.295	0.342	0.340	0.341	0.341	0.323	0.287	0.273	0.247	0.224		
	taud		2.299	2.255	2.195	2.240	2.308	2.399	2.425	2.447	2.495	2.448	2.351	2.288		
	Ebn at Noon		268	287	288	285	290	290	288	289	291	275	255	249		
	Edh at Noon		20	28	36	40	39	36	34	32	27	23	19	18		
(44) All-Sky Solar Radiation	RadAvg		436	753	1244	1689	1988	2006	1897	1691	1338	841	444	331		
	RadStd		18	24	53	56	124	106	115	98	76	62	30	15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-4.27	N/A	-1.37	-2.00	N/A	N/A	N/A	N/A

Nomenclature: See separate page